

Functional Design Document
Life Science Intelligence Center

1. Document Control

Item	Description
Document name	Functional Design Document
Project	Life Science Intelligence Center
Business domain	Clinical Trials and Life Sciences Analytics
Version	1.0
Status	Draft for review
Primary owner	Product Owner / Clinical Analytics Lead
Technical owner	Data Engineering Lead
Business approvers	Clinical Operations, Research, Safety, Claims, Data Governance
Related document	Technical Design Document
Target platform	Databricks Lakehouse on AWS
Consumption layer	Power BI and approved analytical interfaces

2. Purpose

The purpose of the Life Science Intelligence Center is to provide a unified analytics platform that integrates patient, visit, clinical trial, site, enrollment, laboratory, biomarker, adverse-event and claims information.

The platform shall support:

1. A consolidated Patient 360 view.
2. Clinical trial enrollment and site-performance monitoring.
3. Patient cohort identification.
4. Laboratory and biomarker analysis.
5. Safety and adverse-event monitoring.
6. Claims and payment-status analytics.
7. Executive and operational dashboards.
8. Data-quality monitoring and exception management.
9. Governed access to sensitive clinical information.
10. Traceable and auditable reporting.

The original functional design identifies Patient 360, enrollment monitoring, biomarker analysis, relationship exploration, dashboards, security and reporting as the core business capabilities.

3. Business Background

Clinical research data is distributed across Electronic Health Record systems, Clinical Trial Management systems, laboratory systems, biomarker systems, safety systems and claims systems.

When these systems operate independently, users face several problems:

- Patient data must be manually reconciled.
- Trial teams lack timely enrollment visibility.
- Site performance is difficult to compare.
- Biomarker measurements cannot easily be connected to patient and trial context.
- Adverse events are monitored separately from enrollment and biomarker data.
- Claims activity is disconnected from the clinical journey.
- Analysts spend significant time preparing and reconciling data.
- Business users receive inconsistent versions of metrics.

The Life Science Intelligence Center addresses these problems by creating a governed and standardized analytical view across all source domains. The project presentation identifies data unification, accelerated trial enrollment and biomarker analysis as its central goals.

4. Business Problems

4.1 Fragmented Patient Information

Patient demographic, diagnosis, visit, enrollment, laboratory, biomarker, adverse-event and claim information exists in separate datasets.

Business impact

- No single patient profile.
- Repeated manual data preparation.
- Difficult patient-journey analysis.
- Increased risk of inconsistent counts.
- Limited ability to compare clinical and financial activity.

4.2 Slow Clinical Trial Enrollment Analysis

Clinical Operations teams require trial, site and enrollment data to understand recruitment progress.

Business impact

- Delayed identification of candidate cohorts.
- Limited visibility into enrollment status.

- Slow comparison of site performance.
- Increased effort to prepare enrollment reports.
- Delayed escalation of underperforming sites.

4.3 Biomarker Data Isolated from Clinical Context

Biomarker measurements must be interpreted with patient, trial and diagnosis information.

Business impact

- Slow marker-level analysis.
- Limited patient stratification.
- Difficulty comparing biomarker expression by trial.
- Inability to calculate responder outcomes without additional response data.

4.4 Safety Information Is Not Integrated

Adverse events must be reviewed alongside patient and trial context.

Business impact

- Delayed identification of severe-event patterns.
- Limited trial-level safety comparison.
- Manual safety report preparation.
- Incomplete longitudinal safety analysis when event dates are unavailable.

4.5 Limited Data Quality Transparency

Business users need visibility into missing, invalid, duplicate and inconsistent records.

Business impact

- Reports may use unreliable data.
- Exceptions are resolved manually.
- No common data-quality score.
- Root-cause analysis takes longer.

5. Business Objectives and Success Measures

Objective	Functional result	Target outcome
Unify clinical data	Common patient, trial, site and event model	Single analytical source of truth
Improve patient identification	Searchable and filterable patient cohorts	45% faster identification
Improve enrollment analytics	Automated enrollment and site dashboards	70% reduction in analysis time
Accelerate biomarker analysis	Marker, patient and trial-level views	40% faster analysis
Improve data quality	Validation, exception and stewardship process	35% improvement
Improve processing efficiency	Scheduled and incremental data availability	60% reduction in pipeline time
Enable scalable analytics	Support production-scale clinical data	500M+ records
Improve governance	Controlled, audited and traceable access	Approved role-based access
Improve decision support	Executive and operational dashboards	Faster data-driven decisions

6. Scope

6.1 In Scope

Patient Analytics

- Patient demographic profile
- Age and age-group analysis
- Diagnosis analysis
- Patient-to-site association
- Visit history
- Enrollment history
- Laboratory history
- Biomarker measurements
- Adverse-event history
- Claims history
- Patient 360 consolidated view

Clinical Trial Analytics

- Trial master information

- Trial phase analysis
- Therapy-area analysis
- Site participation
- Patient enrollment
- Enrollment status monitoring
- Trial-level patient counts
- Site-level enrollment performance
- Withdrawn and completed enrollment analysis

Laboratory Analytics

- Laboratory test history
- Test-level result analysis
- Patient-level test analysis
- Visit-linked laboratory analysis
- Pre-visit and post-visit result identification
- Data-quality exceptions for invalid or inconsistent dates

Biomarker Analytics

- Biomarker measurement tracking
- Biomarker coverage
- Expression-level analysis
- Marker analysis by patient
- Marker analysis by trial
- Marker analysis by diagnosis
- Patient stratification based on approved marker rules

Safety Analytics

- Adverse-event counts
- Event-type analysis
- Severity analysis
- Trial-level adverse-event analysis
- Patient-level adverse-event history
- Severe-event monitoring
- Batch notification for approved safety thresholds

Claims Analytics

- Claim amount analysis
- Claim-status analysis
- Patient-level claims
- Visit-linked claims
- Approved, pending and rejected claims
- Total and average claim amounts

Reporting and Governance

- Executive dashboards
- Patient 360 dashboard
- Trial and site dashboard
- Biomarker dashboard
- Safety dashboard
- Claims dashboard
- Data-quality dashboard
- Scheduled operational reports
- Role-based access
- Data masking
- Auditability and lineage

The original FDD includes patient, clinical trial, biomarker, safety and claims analytics within scope.

6.2 Out of Scope

The following items are excluded from the current production baseline unless separately approved:

- Real-time streaming analytics
- Direct clinical diagnosis or treatment recommendation
- Automated final patient-eligibility decisions
- Source-system write-back
- Clinical trial protocol authoring
- Drug manufacturing analytics
- Financial accounting and general-ledger processing
- Claim adjudication
- Regulatory submission generation
- Automated pharmacovigilance case submission
- Medical image analysis
- Genomic sequence processing
- Automated machine-learning treatment recommendations
- Knowledge graph implementation, unless approved as a separate phase

6.3 Conditional Scope

The following functions require additional source data or approved business rules:

- True trial enrollment rate against target
- Enrollment velocity against planned schedule
- Time from eligibility to enrollment
- Patient response and responder classification
- Normal/abnormal laboratory interpretation
- Adverse-event chronology
- Seriousness and expectedness analysis
- Active-trial reporting
- Claim-date trend reporting
- Provider and payer analysis

7. Stakeholders and Responsibilities

Stakeholder	Primary responsibility
Executive Leadership	Review enterprise KPIs and business outcomes
Clinical Operations	Monitor trial enrollment and site performance
Clinical Researchers	Explore patient, visit, trial and laboratory data
Biomarker Researchers	Analyze biomarker measurements and patient segments
Safety Team	Monitor adverse-event frequency and severity
Claims Analysts	Analyze claims amounts and statuses
Data Analysts	Perform approved self-service analysis
Data Stewards	Review data quality, mappings and reference values
Product Owner	Own priorities, requirements and acceptance
Data Engineering	Deliver and operate data pipelines
BI Team	Build dashboards and semantic models
Security Team	Approve roles, masking and access controls
QA/UAT Team	Validate functional requirements and business outputs

8. Source Data Inventory

The current functional baseline uses nine source datasets.

Source dataset	Current sample rows	Business entity	Main functional use
ehr_patient_raw	500	Patient	Patient profile and segmentation
ehr_visit_raw	500	Visit	Patient journey and visit analytics
trial_raw	30	Trial	Trial phase and therapy-area analysis
site_raw	50	Site	Site status and country analysis
enrollment_raw	500	Enrollment	Trial and site enrollment monitoring
lab_result_raw	500	Lab result	Test-result and visit analysis
biomarker_raw	500	Biomarker measurement	Marker expression analysis
adverse_event_raw	500	Adverse event	Safety and severity analysis
claims_raw	500	Claim	Claim amount and status analysis

9. Source-to-Functional Mapping

9.1 Patient Data Mapping

Source: ehr_patient_raw

Source field	Functional meaning	Functional rule	Business use
patient_id	Unique patient identifier	Mandatory and unique	Patient 360 joining key
gender	Recorded gender code	Map through approved reference list	Demographic analysis
age	Patient age	Numeric; validate approved range	Age segmentation
diagnosis	Primary recorded diagnosis	Standardize against diagnosis reference	Diagnosis cohorts
site_id	Associated clinical site	Must reference a valid site	Site-based filtering

Derived attributes

Attribute	Proposed rule
Age group	Configurable business bands
Patient source coverage	Indicates which related datasets contain the patient
Has visit	Yes when at least one visit exists
Has enrollment	Yes when at least one enrollment exists
Has lab result	Yes when at least one lab record exists
Has biomarker	Yes when at least one biomarker measurement exists
Has adverse event	Yes when at least one adverse event exists
Has claim	Yes when at least one claim exists

Limitation

The source contains age rather than date of birth. Age may become outdated unless refreshed by the source. Patient name, address and contact information are not present, which supports pseudonymous analytics but limits operational contact workflows.

9.2 Visit Data Mapping

Source: `ehr_visit_raw`

Source field	Functional meaning	Functional rule	Business use
<code>visit_id</code>	Unique visit identifier	Mandatory and unique	Visit-level analysis
<code>patient_id</code>	Patient attending visit	Must exist in patient data	Patient journey
<code>visit_date</code>	Date of visit	Convert to standard date	Timeline and trends
<code>visit_type</code>	Type of visit	Approved values: Emergency, Follow-up, Routine, Screening	Visit segmentation
<code>site_id</code>	Site where visit occurred	Must exist in site data	Site analytics

Functional outputs

- Chronological visit timeline
- Visits per patient
- Visits by type
- Visits by site
- Visits by diagnosis
- Visit trends over time

9.3 Trial Data Mapping

Source: `trial_raw`

Source field	Functional meaning	Functional rule	Business use
<code>trial_id</code>	Unique clinical trial identifier	Mandatory and unique	Trial joins
<code>trial_name</code>	Trial display name	Mandatory	Dashboard display
<code>phase</code>	Clinical trial phase	Approved values I, II, III, IV	Phase analysis
<code>therapy_area</code>	Trial therapeutic area	Standardized reference value	Therapy-area analysis

Current functional limitation

The dataset does not contain:

- Trial status
- Start date
- End date
- Planned enrollment target
- Sponsor
- Protocol identifier
- Eligibility criteria
- Primary endpoint

Therefore, **active trial count**, **target enrollment rate**, **schedule variance** and **protocol-based eligibility** cannot be calculated from the current source alone.

9.4 Site Data Mapping

Source: `site_raw`

Source field	Functional meaning	Functional rule	Business use
site_id	Unique site identifier	Mandatory and unique	Site joins
site_name	Site display name	Mandatory	Reporting
country	Site country	Standardized reference value	Geographic analysis
status	Site operational status	Active or Inactive	Site filtering

Business rule

A new enrollment should normally be associated with an active site. Historical enrollment and visit records related to an inactive site must remain available.

9.5 Enrollment Data Mapping

Source: `enrollment_raw`

Source field	Functional meaning	Functional rule	Business use
enrollment_id	Unique enrollment record	Mandatory and unique	Enrollment fact
patient_id	Enrolled patient	Must reference patient	Patient enrollment history
trial_id	Trial assignment	Must reference trial	Trial monitoring

Source field	Functional meaning	Functional rule	Business use
site_id	Enrollment site	Must reference site	Site performance
enrollment_date	Enrollment date	Valid standardized date	Enrollment trend
status	Enrollment state	Enrolled, Completed or Withdrawn	Funnel analysis

Derived attributes

Derived field	Rule
Enrolled flag	1 when status is Enrolled or Completed
Completed flag	1 when status is Completed
Withdrawn flag	1 when status is Withdrawn
Current participation flag	1 when status is Enrolled
Enrollment year/month	Derived from enrollment date

Important interpretation

The current records represent enrollment outcomes or statuses. They do not include screened, eligible or consented stages. Therefore, the platform can provide a status distribution but not a complete recruitment funnel unless additional stages are provided.

9.6 Laboratory Result Mapping

Source: lab_result_raw

Source field	Functional meaning	Functional rule	Business use
lab_result_id	Unique laboratory result	Mandatory and unique	Laboratory event
visit_id	Related visit	Must reference visit	Visit-lab linkage
patient_id	Related patient	Must match visit patient	Patient history
test_name	Laboratory test	Approved values currently include CBC, CRP, HbA1c and LFT	Test analysis
result_value	Numeric test result	Must be numeric	Result analysis
result_date	Test result date	Valid standardized date	Timeline analysis

Derived fields

Derived field	Rule
Pre-visit result flag	Result date earlier than visit date
Same-day result flag	Result date equals visit date
Post-visit result flag	Result date later than visit date

Data-quality observation

In the uploaded sample, **239 of 500 laboratory results occur before the linked visit date.** This may represent valid pre-visit or screening testing, but the source does not provide test context. The system shall treat this as a warning rather than automatically rejecting the result until the business confirms the rule.

Limitation

The source does not provide units, reference ranges, specimen type or abnormal-result flags. The platform must not classify a result as clinically normal or abnormal until those fields or approved test-specific rules are supplied.

9.7 Biomarker Mapping

Source: `biomarker_raw`

Source field	Functional meaning	Functional rule	Business use
<code>biomarker_id</code>	Unique biomarker measurement record	Mandatory and unique	Measurement-level analysis
<code>patient_id</code>	Related patient	Must reference patient	Patient stratification
<code>trial_id</code>	Related trial	Must reference trial	Trial-marker analysis
<code>marker_name</code>	Biomarker name	Standardized marker reference	Marker grouping
<code>expression_level</code>	Recorded expression value	Numeric and range-validated after scale approval	Expression analysis

Current marker values include BRCA, EGFR, HER2 and PD-L1.

Functional modeling rule

biomarker_raw represents a **measurement or observation**, not only a biomarker master table. The functional model shall separate:

- Biomarker reference information, such as marker name.
- Patient/trial-specific biomarker measurements.

Limitation

The source does not include:

- Measurement date
- Unit or scale
- Test method
- Positive/negative result
- Threshold
- Outcome or treatment response

Therefore, a responder flag cannot be generated reliably from expression level alone.

9.8 Adverse Event Mapping

Source: adverse_event_raw

Source field	Functional meaning	Functional rule	Business use
ae_id	Unique adverse-event record	Mandatory and unique	Safety event
patient_id	Affected patient	Must reference patient	Patient safety history
trial_id	Related trial	Must reference trial	Trial safety analysis
event_type	Event description/category	Standardized reference value	Event grouping
severity	Severity level	Mild, Moderate or Severe	Severity analysis

Derived analytical score

Severity Analytical score

Mild 1

Moderate 2

Severe 3

The score is intended only for reporting and aggregation. It is not a replacement for a clinically approved safety classification.

9.9 Claims Mapping

Source: `claims_raw`

Source field	Functional meaning	Functional rule	Business use
<code>claim_id</code>	Unique claim identifier	Mandatory and unique	Claims analysis
<code>patient_id</code>	Related patient	Must reference patient	Patient financial history
<code>visit_id</code>	Related visit	Must reference visit	Visit-claim linkage
<code>claim_amount</code>	Claim monetary value	Must be numeric and greater than zero	Financial analysis
<code>claim_status</code>	Processing status	Approved, Pending or Rejected	Status analysis

Derived fields

- Approved claim flag
- Pending claim flag
- Rejected claim flag
- Claim amount band
- Visit date used as analytical service date only when business-approved

10. Logical Business Relationships

Site

- has Patients
- hosts Visits
- manages Enrollments

Patient

- attends Visits
- enrolls in Trials through Sites
- receives Lab Results
- has Biomarker Measurements
- experiences Adverse Events
- has Claims linked to Visits

Trial

- contains Enrollments
- has Biomarker Measurements
- has Adverse Events

Laboratory tests and biomarkers must remain separate functional domains. A laboratory test such as CBC or LFT must not be treated as a biomarker such as EGFR or HER2.

11. Functional Process Flows

11.1 Data Availability and Validation Process

```
Source data received
    ↓
Source completeness validation
    ↓
Key and relationship validation
    ↓
Business-rule validation
    ↓
Valid records made available
    ↓
Invalid records placed in exception view
    ↓
Data Steward review and resolution
    ↓
Dashboard and report refresh
```

Functional expectations

- Every processing run must have a status.
- Source and accepted record counts must be visible.
- Rejected and warning counts must be visible.
- Business users must be able to distinguish incomplete data from zero activity.
- Corrected data must be included in the next approved refresh or controlled reprocessing.

11.2 Patient 360 Process

```
Patient master
+ Visits
+ Enrollments
+ Lab results
+ Biomarker measurements
+ Adverse events
+ Claims
    ↓
Unified patient profile
    ↓
Timeline, indicators and drill-through
```

Expected output

The Patient 360 screen shall display:

- Pseudonymous patient identifier
- Gender
- Age and age group
- Diagnosis
- Associated site
- Visit history
- Trial enrollment history
- Laboratory history
- Biomarker measurements
- Adverse-event history
- Claims history
- Data-source coverage indicators

11.3 Enrollment Monitoring Process

```
Trial master
+ Site master
+ Enrollment records
+ Patient attributes
    ↓
Trial and site enrollment metrics
    ↓
Filters and drill-through
    ↓
Operational follow-up
```

Expected output

- Enrollment count by trial
- Enrollment count by site
- Enrollment count by country
- Current enrolled count
- Completed count
- Withdrawn count
- Status distribution
- Monthly enrollment trend
- Diagnosis and demographic breakdown

11.4 Cohort Identification Process

```
User selects criteria
    ↓
Diagnosis, age, gender, site, trial, marker and visit filters applied
    ↓
Candidate cohort returned
    ↓
```

User reviews de-identified patient-level details
↓
Authorized user exports or hands off candidate list

Important control

The system provides analytical candidate identification only. Final eligibility must be determined by qualified clinical staff using the approved protocol.

11.5 Biomarker Analysis Process

Biomarker measurements
+ Patient attributes
+ Trial attributes
+ Diagnosis
↓
Marker-level and cohort-level analysis
↓
Patient stratification

Responder versus non-responder analysis is conditional on the addition of an approved outcome source and response rule.

11.6 Safety Monitoring Process

Adverse events
+ Patient
+ Trial
↓
Event type and severity analysis
↓
Severe-event review queue
↓
Authorized safety-team review

The current process is analytical and batch-based. It is not a replacement for a validated safety case-management system.

11.7 Claims Analysis Process

Claims
+ Patient
+ Visit
↓
Claim status and amount analysis
↓
Patient, visit and site drill-through

12. Detailed Core Requirements

FR-008 Unified Patient 360 View

Description

The system shall consolidate the patient's demographic, diagnosis, site, visit, enrollment, laboratory, biomarker, adverse-event and claim information into one profile.

Inputs

- Patient
- Visit
- Enrollment
- Trial
- Site
- Laboratory result
- Biomarker measurement
- Adverse event
- Claim

Users

- Clinical Researchers
- Clinical Operations
- Safety Team
- Authorized Data Analysts

Business rules

1. Patient ID is the primary linking identifier.
2. Related information shall be displayed only when the relationship is valid.
3. Missing related data shall display as "No data available," not zero.
4. Patient records shall be masked or pseudonymized according to the user's role.
5. Events shall be ordered chronologically where a date is available.
6. Undated events, such as current adverse events, shall appear in a separate section.

Acceptance criteria

- A user can search for an authorized patient ID.
- The correct demographic and diagnosis details are displayed.
- All linked visits, enrollments, labs, biomarkers, adverse events and claims are available.
- Invalid cross-patient records are excluded and listed as data-quality exceptions.
- Unauthorized users cannot access patient-level details.
- The user can drill from summary indicators to individual records.

FR-015 Clinical Trial Enrollment Monitoring

Description

The system shall provide enrollment visibility by trial, site, country, status and enrollment date.

Inputs

- Trial
- Site
- Enrollment
- Patient

Outputs

- Total enrollment records
- Currently enrolled patients
- Completed patients
- Withdrawn patients
- Enrollment trend
- Enrollment by phase
- Enrollment by therapy area
- Enrollment by site and country

Business rules

- Enrollment counts shall use distinct enrollment IDs.
- Patient counts shall use distinct patient IDs.
- Status values outside the approved list shall be treated as exceptions.
- Historical enrollment at inactive sites shall remain visible.
- True target-based enrollment rate shall not be displayed until target data is available.

Acceptance criteria

- Trial and site totals reconcile to accepted enrollment records.
- Users can filter by trial, phase, therapy area, site, country, date and status.
- Users can drill from trial to site to patient enrollment.
- Withdrawn records are not counted as currently enrolled.
- Completed records are counted separately from currently enrolled records.

FR-018 Patient Cohort Identification

Description

The system shall allow authorized users to identify analytical patient cohorts using available clinical criteria.

Supported criteria

- Age or age group
- Gender
- Diagnosis
- Site
- Country
- Visit type
- Trial
- Trial phase
- Therapy area
- Enrollment status
- Laboratory test
- Biomarker name
- Biomarker expression range
- Adverse-event type
- Adverse-event severity

Rules

- The result is an analytical candidate cohort, not a final eligibility decision.
- Every result must show which filters selected the patient.
- The platform must avoid duplicate patients in cohort counts.
- Export requires an authorized role.
- Protocol-based inclusion/exclusion criteria require a separate rules source.

Acceptance criteria

- Applying the same filters repeatedly returns the same approved dataset version.
- Cohort count equals the number of distinct patients.
- Users can save approved filter combinations.
- Exported cohorts include an extraction timestamp and filter summary.

FR-020 Laboratory Result Analysis

Description

The system shall provide laboratory-result analysis by patient, visit, test, date and site.

Rules

- Result value must be numeric.
- Test names must use an approved reference list.
- Patient ID must match the patient linked to the visit.

- Pre-visit results shall be flagged, not rejected automatically.
- Normal/abnormal classification shall remain unavailable until units and reference ranges are supplied.

Acceptance criteria

- Users can view all accepted lab results for a patient.
- Users can filter by test name and result date.
- Pre-visit results are clearly identified.
- Invalid visit or patient relationships are shown in the exception dashboard.
- The system does not label results normal or abnormal without approved clinical ranges.

FR-022 Biomarker Measurement Analysis

Description

The system shall analyze biomarker expression by marker, patient, trial, diagnosis and therapy area.

Rules

- Marker names must be standardized.
- Expression level must be numeric.
- Expression thresholds must be configurable and business-approved.
- Marker measurement and marker reference information must be treated separately.
- No responder classification may be inferred from expression alone.

Acceptance criteria

- Users can compare average, minimum, maximum and distribution of expression values.
- Users can filter by marker, trial, phase, therapy area and diagnosis.
- Patient-level drill-through is available to authorized users.
- A missing unit or threshold is visibly indicated.
- Responder views remain disabled until response data is approved.

FR-026 Adverse-Event Monitoring

Description

The system shall provide adverse-event analysis by event type, severity, patient and trial.

Rules

- Severity values must be Mild, Moderate or Severe.
- Severe events shall be highlighted.
- The severity score is for analytics only.
- Event trend charts require an event date.
- Undated adverse events shall not be assigned a synthetic clinical date.

Acceptance criteria

- Event totals reconcile to accepted adverse-event records.
- Users can filter by trial, event type and severity.
- Severe events are available in a dedicated review list.
- Users can drill to patient and trial context when authorized.
- No time-to-onset or event-trend calculation is shown without an event date.

FR-030 — Claims Analytics

Description

The system shall provide claim amount and claim-status analytics linked to patients and visits.

Rules

- Claim amount must be greater than zero.
- Claim status must be Approved, Pending or Rejected.
- Patient on the claim must match the patient on the linked visit.
- Visit date may not be represented as claim date unless formally approved.
- Currency must be displayed as unspecified until a currency field or business rule is supplied.

Acceptance criteria

- Claim totals reconcile to accepted claim records.
- Users can filter by claim status, patient, visit and site.
- Approved, pending and rejected counts are shown separately.
- Total and average claim amounts are calculated correctly.
- Reports clearly indicate when claim date or currency is unavailable.

13. Dashboard Requirements

13.1 Executive Dashboard

Users

Executive Leadership, Program Directors and Product Owners.

KPIs

- Total patients
- Total trials
- Active sites
- Total enrollments
- Enrollment-status distribution
- Biomarker coverage
- Severe adverse events
- Total claim amount
- Data-quality pass percentage
- Latest successful refresh time

Filters

- Date
- Country
- Site
- Trial
- Trial phase
- Therapy area
- Diagnosis

13.2 Patient 360 Dashboard

Sections

1. Patient summary
2. Visit timeline
3. Enrollment history
4. Laboratory results
5. Biomarker measurements
6. Adverse events
7. Claims
8. Data-source coverage

Controls

- Patient search
- Date range
- Trial filter
- Event-type filter
- Download restricted by role

13.3 Clinical Trial Dashboard

KPIs

- Total enrollments
- Current enrolled
- Completed
- Withdrawn
- Enrollment trend
- Enrollment by trial
- Enrollment by phase
- Enrollment by therapy area
- Site ranking
- Country distribution

Drill path

Therapy Area → Trial → Site → Enrollment → Patient

13.4 Biomarker Dashboard

KPIs and visuals

- Patients with biomarker measurements
- Biomarker coverage
- Expression-level distribution
- Average expression by marker
- Marker by diagnosis
- Marker by trial
- Marker by therapy area
- Patient cohort table

Responder versus non-responder sections shall be disabled until approved response data is available.

13.5 Safety Dashboard

KPIs

- Total adverse events
- Events by type
- Events by severity
- Severe-event percentage
- Events by trial
- Events by diagnosis

- Severe-event patient list

Time trends shall be omitted until event dates become available.

13.6 Claims Dashboard

KPIs

- Total claims
- Total claim amount
- Average claim amount
- Approved claims
- Pending claims
- Rejected claims
- Claim amount by status
- Claims by patient and site

Date trends shall require claim or service date.

13.7 Data Quality Dashboard

KPIs

- Source record count
- Accepted record count
- Rejected record count
- Warning count
- Duplicate count
- Missing mandatory field count
- Invalid relationship count
- Pre-visit lab-result warning count
- Latest processing status
- Data-quality trend by source